
Exercises

Advanced Lesson 10

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 - Advanced Mathematical Modeling

1: Construct a Module which will:

Generate a list of 10 integers, $1 \leq i \leq 10$

Average the numbers in the list

Return the average

```
Clear[m1];  
m1 := Module[{list}, list = Table[i, {i, 1, 10}]; Mean[list]]
```

m1

2: Construct a Module which will:

Generate a list of 10 ordered pairs $\{i, i^2\}$, $1 \leq i \leq 10$

Return the plot of the list

```
Clear[m2];  
m2 := Module[{pairs}, pairs = Table[{i, i^2}, {i, 1, 10}];  
ListPlot[pairs, PlotStyle -> PointSize[0.02`]]]
```

m2

3: Construct a Module which will:

Define a function: $f[a_, y0_] = \text{Module}[\{\}, \dots]$ which will:

Solve the ODE: $y'[t] = a*y[t]$

Subject to the initial condition: $y[0] = y0$

Plot $y[t]$ for $0 \leq t \leq 2$

Evaluate: $f[2,3]$ and $f[-2,3]$

```
Clear[f, a, y0, t, y];  
f[a_, y0_] := Module[{sol, yt}, sol = DSolve[{y'[t] == a y[t], y[0] == y0}, y[t], t];  
yt = y[t] /. First[sol];  
Plot[yt, {t, 0, 2}, PlotRange -> All, PlotLabel -> "a = "a", y0 = "y0"]]
```

f[2, 3]

f[-2, 3]

4: Construct a Module exactly as in problem 3, except that $y0$ will have an assigned value of 1.

Define a function: $f[a_] = \text{Module}[\{\text{make changes to } y0 \text{ here}\}, \dots \text{keep the same } \dots]$

Evaluate: $f[2]$ and $f[-2]$

```
Clear[f, a, t, y];
f[a_] := Module[{y0 = 1, sol, yt}, sol = DSolve[{y'[t] == a y[t], y[0] == y0}, y[t], t];
  yt = y[t] /. First[sol];
  Plot[yt, {t, 0, 2}, PlotRange -> All, PlotLabel -> "a = "a", y0 = "y0"]
```

f[2]

f[-2]

5: Construct a Module which will:

Define a function: $f[n_] = \text{Module}[\{\}, \dots]$ which will:

Generate a list of n integers, $1 \leq i \leq n$

Average the numbers in the list

Return the average

Evaluate: $f[10]$ and $f[100]$

```
Clear[f, n];
f[n_] := Module[{list}, list = Table[i, {i, 1, n}]; Mean[list]]
{f[10], f[100]}
```

6: In simple economic theory, the state of the economy in year " n " is a function of the economy in the previous year. If the economic model is linear, then the equation for year " n " is:

$\text{economy}[n_] := a * \text{economy}[n-1] + b$

This defines recursively the state of our economy in terms of previous years.

Assume that at year 1, the initial state of the economy was = value1

Define a function: $\text{economy}[n_] := a * \text{economy}[n-1] + b$

Evaluate: $\text{economy}[1]$, $\text{economy}[2]$, $\text{economy}[10]$

```
Clear[economy, a, b, n, value1];
economy[1] = value1;
economy[n_] := a economy[n - 1] + b
{economy[1], economy[2], economy[10]}
```

7: In simple economic theory, the state of the economy in year " n " is a function of the economy in the previous year. If the economic model is NOT linear, then the equation for year " n " is:

$\text{economy}[n_] := F[\text{economy}[n-1]]$, for some nonlinear function " F "

This defines recursively the state of our economy in terms of previous years.

Assume that at year 1, the initial state of the economy was = value1

Define a function: $\text{economy}[n_] \text{ if } F[X_] := \sqrt{aX + b}$

Evaluate: $\text{economy}[1]$, $\text{economy}[2]$, $\text{economy}[10]$

```
Clear[economy, F, X, a, b, n, value1];
F[X_] := a X + b;
economy[1] = value1;
economy[n_] := F[economy[n - 1]]
{economy[1], economy[2], economy[10]}
```

8: Assume that the state of the economy in year " n " is a function of the economy in the previous two years. Also, if the economic model is NOT linear, then the equation for year " n " is:

$\text{economy}[n_]:=F[\text{economy}[n-1], \text{economy}[n-2]]$, for some nonlinear function "F"

This defines recursively the state of our economy in terms of previous years.

Assume that at year 1, the initial state of the economy was = value1

Assume that at year 2, the initial state of the economy was = value2

Define a function: $\text{economy}[n_]$ if $F[X_ , Y_] := \sqrt{aX + bY}$

Evaluate: $\text{economy}[1]$, $\text{economy}[2]$, $\text{economy}[3]$, $\text{economy}[5]$

```
Clear[economy, F, X, Y, a, b, n, value1, value2];
```

```
F[X_, Y_] := a X + b Y;
```

```
economy[1] = value1;
```

```
economy[2] = value2;
```

```
economy[n_] := F[economy[n - 1], economy[n - 2]]
```

```
{economy[1], economy[2], economy[3], economy[5]}
```

9: Using procedural programming, define the economy in terms of a module (i.e. for $k=1,2, \dots, n$) (Use the information supplied in Problem 6):

$\text{economy}[n_] := \text{Module}[\{\}, \dots \text{using Do} \dots]$

$\text{economy}[n_] := \text{Module}[\{\}, \dots \text{using While} \dots]$

$\text{economy}[n_] := \text{Module}[\{\}, \dots \text{using For} \dots]$

Evaluate: $\text{economy}[1]$, $\text{economy}[2]$, $\text{economy}[10]$

```
Clear[economyDo, economyWhile, economyFor, a, b, n, k, value1];
```

```
economyDo[n_] := Module[{e = value1}, Do[e = a e + b, {k, 2, n}];  
e]
```

```
economyWhile[n_] := Module[{e = value1, k = 1}, While[k < n, e = a e + b; k = k + 1]; e]
```

```
economyFor[n_] := Module[{e = value1, k}, For[k = 2, k ≤ n, k = k + 1, e = a e + b]; e]
```

```
{economyDo[1], economyDo[2], economyDo[10]}
```

```
{economyWhile[1], economyWhile[2], economyWhile[10]}
```

```
{economyFor[1], economyFor[2], economyFor[10]}
```

10: Using the Which command, define a piecewise function such that:

$f(x) = x^2$, for $x < -2$

$f(x) = \sin[x^2]$, for $-2 \leq x < 1$

$f(x) = 2x$, for $1 \leq x < 3$

$f(x) = x$, for $3 \leq x$

Plot the function for: $-3 \leq x < 4$

```
Clear[f, x];
```

```
f[x_] := Which[x < -2, x^2, -2 ≤ x < 1, Sin[x^2], 1 ≤ x < 3, 2 x, 3 ≤ x, x]
```

```
Plot[f[x], {x, -3, 4}]
```